

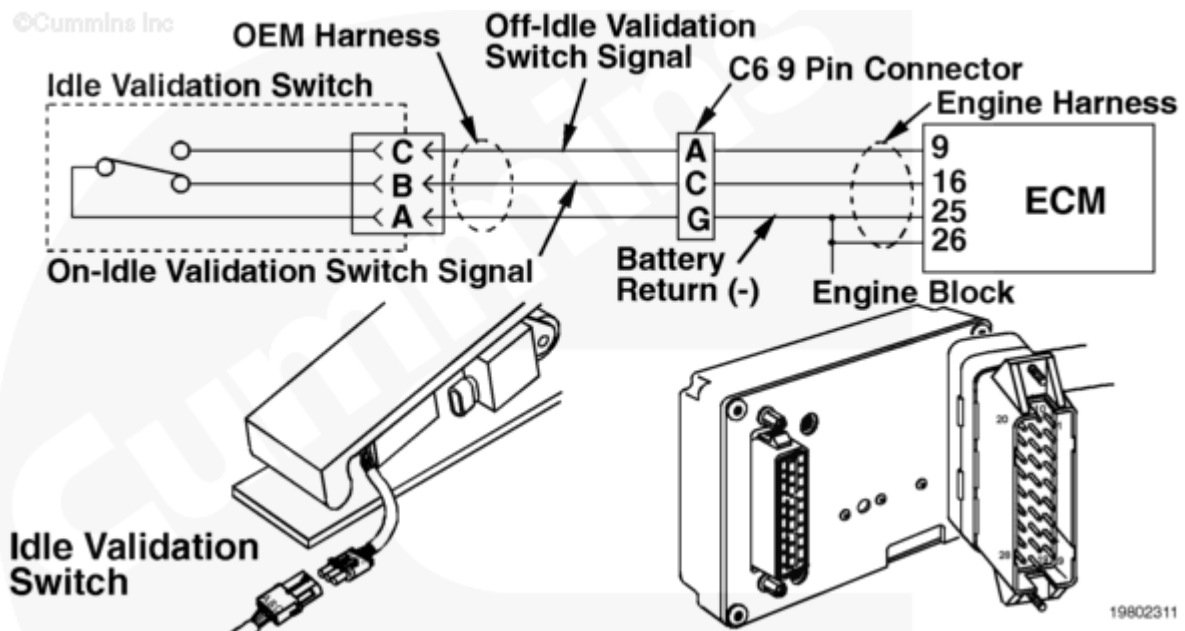
Fault Code: 431

Accelerator Pedal or Lever Idle Validation Switch - Data Erratic, Intermittent, or Incorrect

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 431 PID(P): P91 SPN: FMI: 2 Lamp: On SRT: 00-633	Voltage detected simultaneously on both idle validation and off-idle validation switches.	Possible reduced engine performance.



Idle Validation Circuit

Circuit Description

The idle validation signal is a safety feature that disables throttle control while an improper validation signal is detected by the ECM. The idle validation switch provides an on-idle and off-idle validation signal to the ECM through the OEM wiring harness and engine wiring harness.

Component Location

The idle validation switch is mounted on the throttle pedal. The throttle pedal location varies with each OEM.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected on-idle signal and off-idle signal were both on or both off at the same time.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp immediately when the diagnostic runs and fails.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Malfunctioning or damaged OEM wiring harness.
- Malfunctioning or damaged throttle position sensor.
- Damaged or loose connectors.

Refer to Troubleshooting Fault Code t05-431 (</qs3/pubsys2/xml/en/procedures/98/98-t05-431.html>)

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